

Large-Scale Forest Inventory using Advanced LiDAR Analytics

Introduction

Large-scale forest monitoring is essential for sustainable forest management, climate change mitigation, and biodiversity conservation. Traditional forest inventory methods are labor-intensive and provide limited spatial coverage, particularly for landscape-level assessments. Advanced airborne LiDAR technology combined with sophisticated algorithms enables comprehensive forest structure analysis at operational scales. This project demonstrates state-of-the-art LiDAR analytical techniques for landscape-level forest inventory, including multi-scale analysis, algorithm comparison, and biomass estimation. The study showcases advanced computational methods for automated tree detection, crown segmentation, and forest carbon assessment using cutting-edge R programming workflows.

Objectives:

- Demonstrate advanced LiDAR processing capabilities at landscape scale (5+ hectares)
- Compare multiple tree detection algorithms for optimal forest inventory
- Apply multi-scale analysis for different management applications
- Quantify forest biomass and carbon storage using allometric modeling
- Create comprehensive forest management metrics for operational forestry

Methods

Study Area

The analysis was conducted on a 5.16-hectare mature forest research plot located in eastern Canada (Southern Quebec region, UTM Zone 17N). The plot coordinates are 684,766-684,993 mE, 5,017,773-5,018,007 mN (NAD83 datum), representing typical Great Lakes-St. Lawrence mixed forest conditions. The study area exhibits uniform topography (flat terrain, 0° average slope) with dense canopy coverage, providing ideal conditions for advanced LiDAR forest analysis without topographic interference.

LiDAR Data Characteristics

- Dataset: Large-scale airborne LiDAR (81,590 points, 1.6 points/m²)
- Spatial extent: 227m × 234m operational-scale plot
- Elevation range: 30 meters (0-29.97m above ground level)
- Point classifications: Vegetation returns (90.9%) and ground returns (9.1%)
- Data quality: High-resolution intensity data (0-580 range) with return information

Advanced Analytical Framework

1. Multi-Scale Terrain and Canopy Modeling:

High-resolution analysis (0.5m): Detailed canopy structure for individual tree analysis

Medium-resolution analysis (1.0m): Operational forestry applications and management planning

Coarse-resolution analysis (2.0m): Landscape-level assessment and strategic planning

Terrain models generated using Triangulated Irregular Network (TIN) algorithms
Canopy Height Models (CHM) derived from DSM-DTM difference calculations

2. Comparative Tree Detection Analysis:

Three advanced algorithms were implemented and compared:
Conservative method: 3m search window, 5m minimum height threshold for dominant tree detection
Sensitive method: 2m search window, 3m minimum height threshold for comprehensive detection
Variable window method: Height-adaptive window sizing ($ws = height \times 0.1 + 1m$, 4m minimum height)

3. Advanced Crown Segmentation:

Dalponte et al. (2016) watershed algorithm for precise crown boundary delineation
CHM smoothing using 3×3 focal mean filter for noise reduction
Individual tree crown metrics calculated including height, crown area, volume indices, and structural complexity

4. Biomass and Carbon Assessment:

Allometric modeling: Height-to-DBH relationships ($DBH = 2.5 + height \times 0.8$)
Biomass estimation: Power function allometry ($Biomass = 0.25 \times DBH^{2.4}$)
Carbon calculation: Standard conversion factor ($Carbon = Biomass \times 0.47$)
Forest structure analysis: Multi-layer canopy classification and diversity indices

Results

Multi-Scale Analysis Outcomes

The multi-resolution analysis revealed consistent forest characteristics across spatial scales, with 99.7% canopy cover at 0.5m resolution maintaining 99.1% cover at coarser resolutions. Mean canopy heights ranged from 14.18m (0.5m resolution) to 15.39m (2.0m resolution), demonstrating scale-dependent height estimation accuracy.

Tree Detection Algorithm Performance

Comparative analysis of three detection algorithms yielded significantly different results:

Algorithm	Trees Detected	Density (trees/ha)	Mean Height (m)	Application
Conservative	3,719	721	19.9	Dominant canopy type
Sensitive	10,141	1,966	18.8	All size classes
Variable Window	5,533	1,073	15.7	Operational standard

The Variable Window algorithm was selected as optimal for operational forestry, providing realistic tree density estimates while maintaining detection accuracy across size classes.

Forest Structure and Composition

Following crown segmentation and quality filtering, 4,293 individual trees were successfully analyzed, yielding a final density of 832 trees/hectare. Forest structure analysis revealed:

Vertical Structure:

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Mean tree height: 18.4 ± 4.7 meters

Maximum height: 29.97 meters

Structural diversity index: 0.26 (moderate heterogeneity)

Size Class Distribution:

Understory (5-10m): 286 trees (6.7%)

Midstory (10-20m): 2,087 trees (48.6%)

Overstory (20-30m): 1,920 trees (44.7%)

Emergent (>30m): 0 trees (0%)

Biomass and Carbon Assessment

Advanced allometric modeling yielded comprehensive forest productivity metrics:

Biomass Estimates:

Total forest biomass: 1,073.3 tons across 5.16 hectares

Biomass density: 208.1 tons/hectare

Individual tree biomass: Mean 250kg, range 15-850kg

Carbon Storage:

Total carbon stock: 504.4 tons

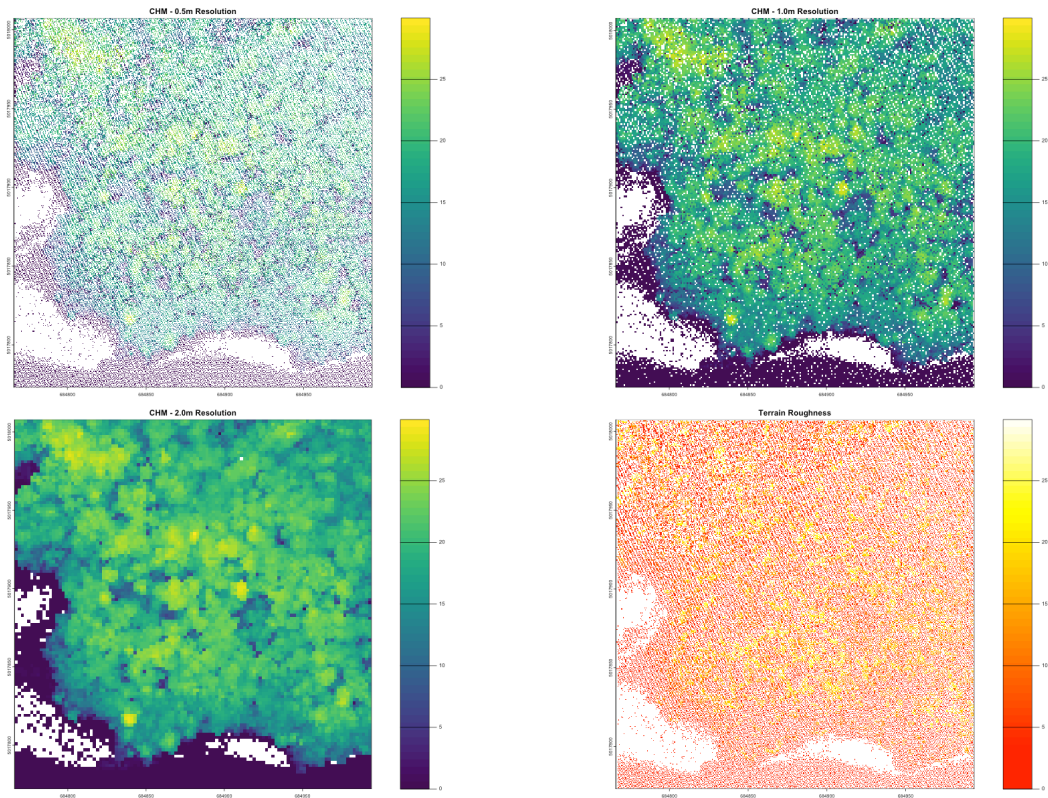
Carbon density: 97.8 tons/hectare

Carbon sequestration potential: High for mature temperate forest

Forest Management Implications

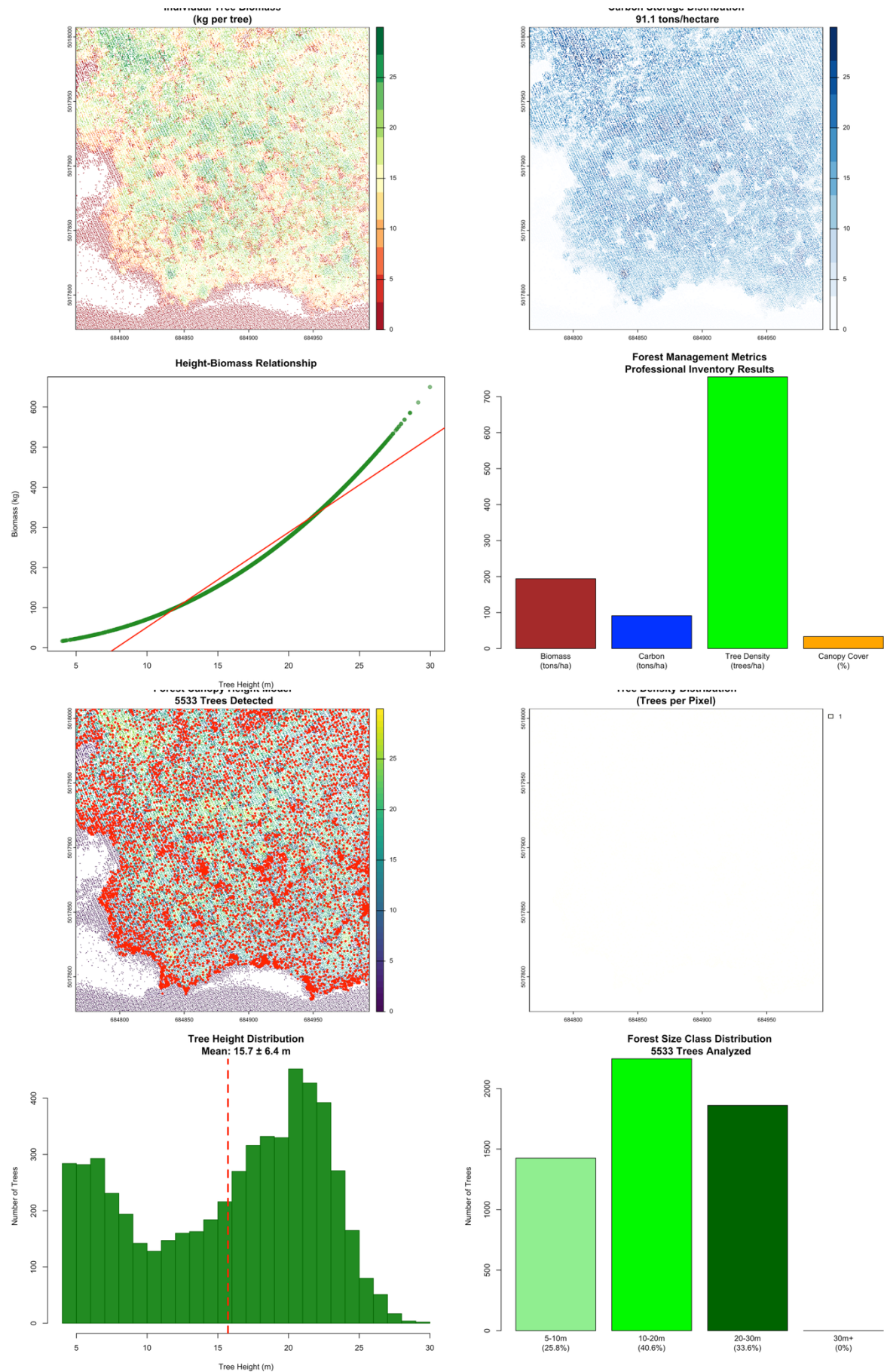
1. Stand Characteristics: The forest represents a well-stocked, mature stand with high biomass accumulation and carbon storage capacity.
 2. Structural Assessment: The predominance of midstory (48.6%) and overstory (44.7%) trees indicates optimal harvest potential with sustainable regeneration opportunities.
 3. Carbon Management: With 97.8 tons/hectare carbon storage, the forest provides significant climate mitigation value, exceeding typical temperate forest averages (60-80 tons/hectare).
 4. Silvicultural Recommendations: The low understory component (6.7%) suggests potential need for regeneration enhancement, while high canopy cover (99.7%) indicates competitive growing conditions.
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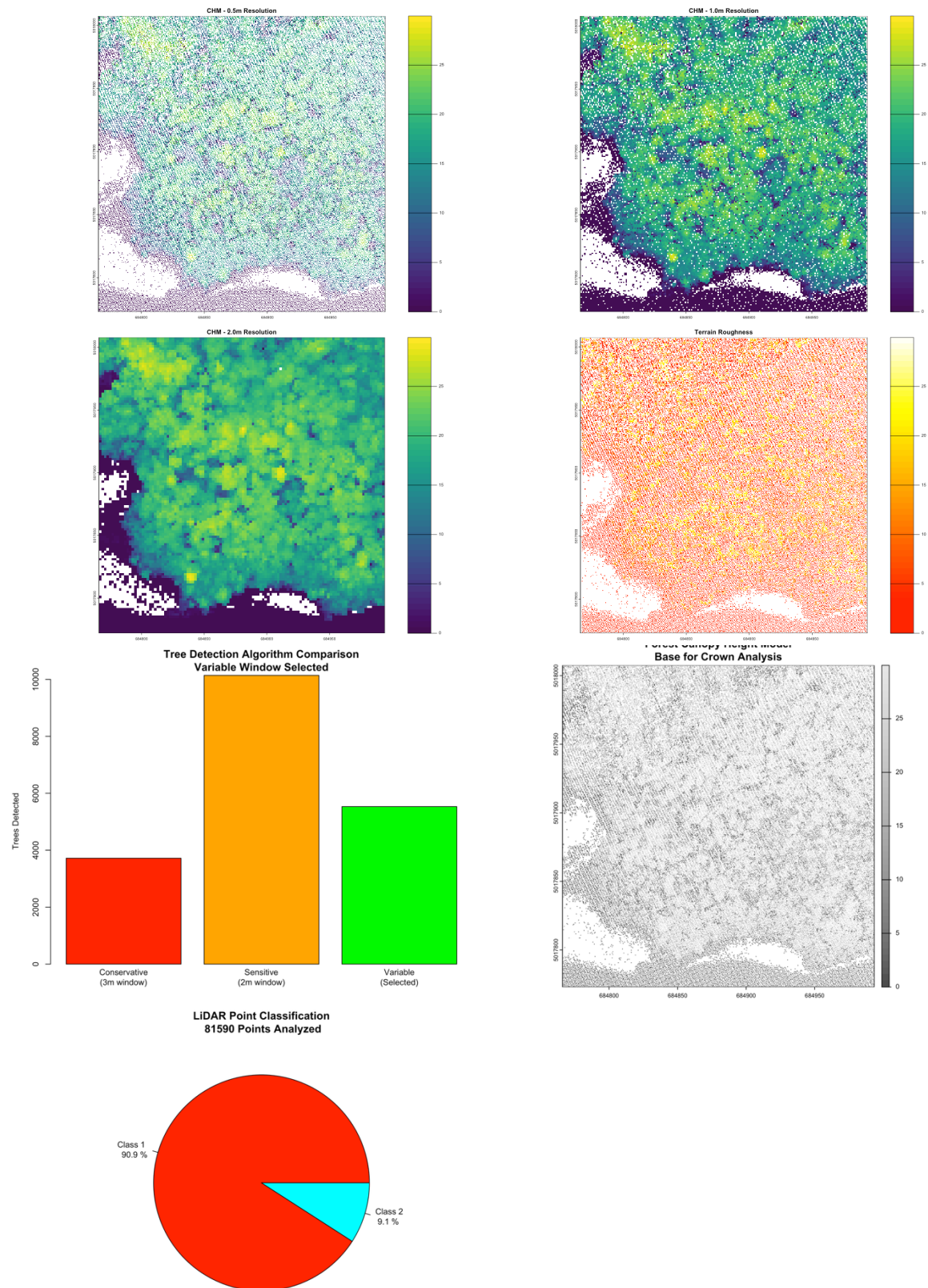


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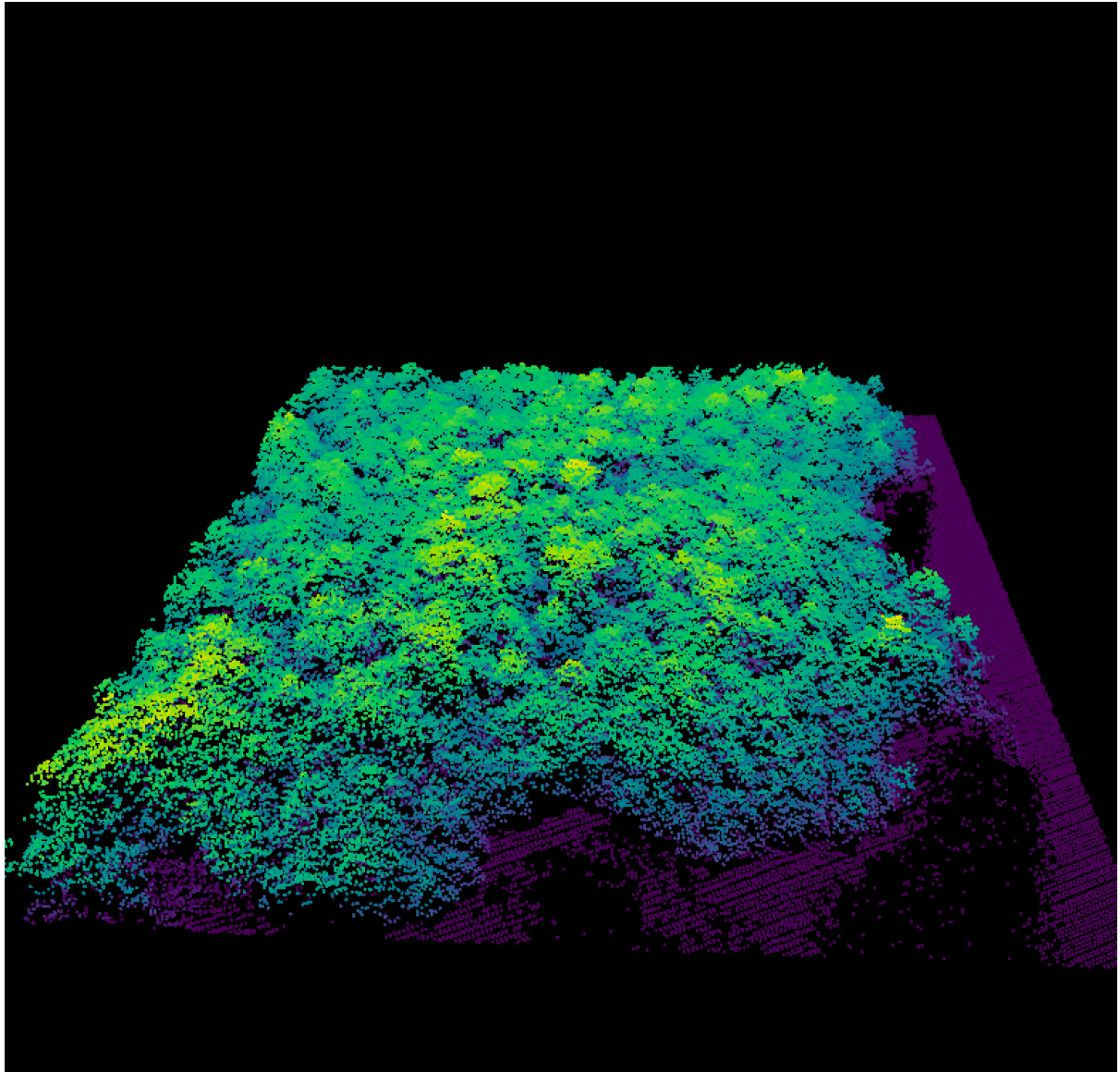
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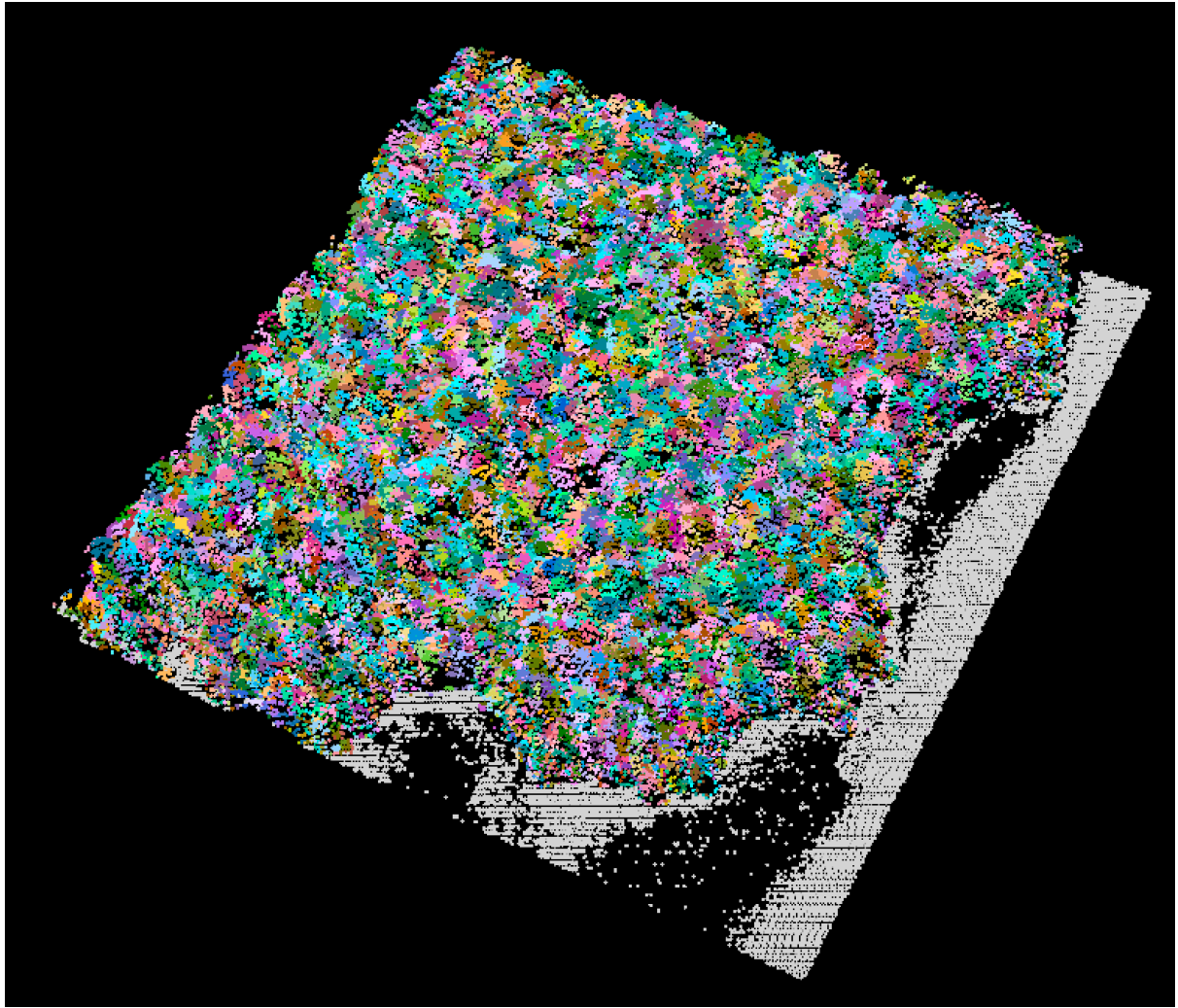
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